智驾伴侣项目知识点介绍

PPT

2435050112

2026 11

PPT vehicleId “ ”

user-service	8081		user_db
vehicle-service	8082		vehicle_db
alert-service	8083		alert_db
driving-behavior-service	8084		driving_db
carbon-service	8085		carbon_db

□□□□□□□ HTTP

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Nacos Docker Compose

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“ ” Repository

“ ” alert_db vehicleId vehicle_db

ID	vehicleId
Repository	Spring Data JPA

HTTP

Nacos

IP vehicle-service Nacos

COURSE_GROUP Nacos

	Nacos 3.2.4	
	COURSE_GROUP	
	18080	
HTTP	8848	Nacos HTTP API
SDK gRPC	9848	SDK

Nacos Nacos verify-discovery “
”

HTTP PI

API HTTP API GET POST GET /api/vehicles/1
1 POST /api/alerts

JSON {"vehicleId":1,"level":2} 1 2 JSON
Java

200		
201		
400		1 3
404		
503		

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Spring Boot

Spring Boot Java Web Spring Boot

Controller Service Repository Controller HTTP Service Repository Entity Java DTO

Controller	HTTP	AlertController
Service		AlertService
Repository		AlertRepository
Entity		Vehicle Alert DrivingEvent
DTO		AlertRequest CarbonReportRequest

JPA Hibernate Java H2 MySQL

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RestTemplate Spring HTTP DiscoveryClient Nacos
RestTemplate

OpenFeign HTTP Java @FeignClient
OpenFeign vehicle-service

Fallback

LoadBalancer	
Fallback	
Circuit Breaker	

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level 1 3 vehicleId pending
alert_db

400 400 503
“ ”

alert-service OpenFeign DiscoveryClient RestTemplate
Nacos HTTP

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100 2 30 3 30 120 km/h 2 20

score = max(0, 100 - min(30, × 2) - min(30, × 3) - min(20, × 2))

HTTP 2.31 CO_2 0.581 CO_2

CO_2 = × 2.31 + × 0.581

vehicleId + report_month

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Docker

Dockerfile Docker Compose YAML

Image	
Container	
Volume	
Port Mapping	
Environment Variable	
Health Check	

H2 Mock MySQL Nacos

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BFF BFF

API Nacos Config SkyWalking Prometheus Grafana

Seata

Nacos	
	fallback 503
	11

答辩时可以把这套项目概括为： Nacos HTTP

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